

Machine Learning (ML) in Hindi

Week-7 Assignment

Total points: 15

Topics covered:

- Support Vector Machine (SVM) Classifiers

Instructions

- Each question is a 1-point multiple choice question (MCQ) with only one correct answer unless specified otherwise.
- Multiple select questions (MSQ) are 2-point questions with one or more than one answers correct
- For programming questions, use the default parameter settings unless specified otherwise.

1. [MCQ] What is the formula for Polynomial Kernel which is used SVM?

- a) $\mathcal{K}(x_i, x_j) = (x_i^T x_j + r)^d$
- b) $\mathcal{K}(x_i, x_j) = \exp\left[\frac{\|x_i - x_j\|}{2\sigma^2}\right]$
- c) $\mathcal{K}(x_i, x_j) = (x_i^T + x_j)$
- d) None of the above

Answer: a) $\mathcal{K}(x_i, x_j) = (x_i^T x_j + r)^d$

2. [MSQ] Which of the following kernels are supported by sklearn.svm.SVC class?

- a) linear
- b) poly
- c) rbf
- d) sigmoid

Answer a), b), c), d)

3. [MCQ] What are the data points called that help build the maximum margin in SVMs?

- a) Support Vectors
- b) Support Matrix
- c) Support Determinant
- d) Support Gradient

Answer a) Support Vectors

4. [MCQ] What is the primal formulation of soft margin SVM classifier?

a) $\min_{\theta, \xi} \frac{1}{2} \theta^T \theta$, s.t. $(x_i^T \theta + b) \geq 1 \quad \forall i = 1, 2, \dots, n$

b) $\max_{\theta, \xi} \frac{1}{2} \theta^T \theta + C \sum_{i=1}^n \xi_i$, s.t. $(x_i^T \theta + b) \geq 1 - \xi_i \quad \forall i = 1, 2, \dots, n$ and $\xi_i \geq 0 \quad \forall i = 1, 2, \dots, n$

c) $\max_{\theta, \xi} \frac{1}{2} \theta^T \theta - C \sum_{i=1}^n \xi_i$, s.t. $(x_i^T \theta + b) \geq 1 - \xi_i \quad \forall i = 1, 2, \dots, n$

d) $\min_{\theta, \xi} \frac{1}{2} \theta^T \theta + C \sum_{i=1}^n \xi_i$, s.t. $(x_i^T \theta + b) \geq 1 - \xi_i \quad \forall i = 1, 2, \dots, n$ and $\xi_i \geq 0 \quad \forall i = 1, 2, \dots, n$

Answer d) $\min_{\theta, \xi} \frac{1}{2} \theta^T \theta + C \sum_{i=1}^n \xi_i$, s.t. $(x_i^T \theta + b) \geq 1 - \xi_i \quad \forall i = 1, 2, \dots, n$ and $\xi_i \geq 0 \quad \forall i = 1, 2, \dots, n$

5. [MCQ] What is a decision hyperplane in the context of SVM?

a) A line that maximises the margin between two classes

b) A geometric plane of n-1 dimensions in an n-dimensional space that separates the two classes

c) A loss function used in SVM

d) A type of Machine Learning Model

Answer b) A geometric plane of n-1 dimensions in an n-dimensional space that separates the two classes

6. [MCQ] What is the primary objective of a Support Vector Machine (SVM) algorithm?

a) To create a decision tree by recursively splitting the data.

b) To find a hyperplane that best separates the data into different classes.

c) To minimize the mean squared error between predicted and actual values.

d) To cluster the data into a predefined number of groups.

Answer b) To find a hyperplane that best separates the data into different classes.

7. [MCQ] In the context of SVM, what are 'support vectors'?

a) The data points that are misclassified by the SVM model.

b) The data points that lie exactly on the hyperplane.

c) All the data points in the training set.

d) The data points that are closest to the hyperplane and influence its position and orientation.

Answer d) The data points that are closest to the hyperplane and influence its position and orientation.

8. [MCQ] What is the purpose of the 'Kernel Trick' in SVMs?

- a) To enable SVMs to perform non-linear classification.
- b) To reduce the number of support vectors.
- c) To automatically select the best features for the model.
- d) To speed up the training process of a linear SVM.

Answer a) To enable SVMs to perform non-linear classification.

9. [MCQ] Which of the following problems is SVM NOT well-suited for?

- a) Text classification problems.
- b) Image classification.
- c) Classification with very large datasets (e.g., millions of samples).
- d) Bioinformatics, like gene classification.

Answer c) Classification with very large datasets (e.g., millions of samples).

10. [MCQ] What is the role of the 'degree' parameter in a polynomial kernel SVM?

- a) It determines the number of support vectors.
- b) It controls the flexibility of the decision boundary.
- c) It determines the number of support vectors.
- d) It sets the penalty for misclassification.

Answer b) It controls the flexibility of the decision boundary.

11. [MCQ] Why is scaling features (e.g., using StandardScaler) important before training an SVM?

- a) To prevent features with larger ranges from dominating the model.
- b) To reduce the number of support vectors.
- c) To convert the problem from non-linear to linear.
- d) To automatically select the kernel function.

Answer a) To prevent features with larger ranges from dominating the model.

12. [MCQ] If two different SVM models for the same dataset have the exact same decision boundary, which statement MUST be true?

- a) They were trained on the exact same number of data points.
- b) Both models are linear SVMs.
- c) Their support vectors and their corresponding weights are the same.
- d) None of the above

Answer c) Their support vectors and their corresponding weights are the same.

13. [MCQ] If adding more data points to your training set does not change the decision boundary at all, what can you conclude about these new points?
- a) They were all correctly classified and lay inside the margin.
 - b) They all became support vectors.
 - c) They were all correctly classified and lay outside the margin.
 - d) The learning algorithm has converged to a poor solution.

Answer c) They were all correctly classified and lay outside the margin.

14. [MCQ] Which of the following python libraries support SVM classifiers?
- a) Numpy
 - b) Pandas
 - c) Scikit Learn
 - d) Seaborn

Answer c) Scikit Learn.